

Perquinolone Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-625/C

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

3.2.P.4 Excipient Control Strategy

Early clinical formulations F-01 through F-03 employed Avicel PH-101. For the commercial formulation F-625/C, Avicel PH-102 (Microcrystalline cellulose, filler-binder) is used.

Formulation Development Rationale

The basis for selection of Avicel PH-102 is superior compactibility at low compression forces, demonstrated across three pilot-scale batches.

The approach described herein is consistent with Ph. Eur. general monograph 2034, ICH Q3D and ICH Q6A.

Commercial Control Strategy — Excipients

The current approved commercial specification (SPEC-EX-4618 Rev 03) lists Microcrystalline cellulose as Avicel PH-101.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Deviations observed during the campaign were investigated and closed with no impact to product quality. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. No changes to the approved analytical procedures are proposed in this submission. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

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